

Release Note for HiSIM_SOTB 1.3.0

Hiroshima University

Revision: HiSIM_SOTB 1.3.0

Date: 2021.11.16

Deliverable List of HiSIM_SOTB 1.3.0

Deliverable	File Name or Directory Name	Date
Source Code (Verilog-A)	HiSIM_SOTB_1.3.0_VA-Code/	2021.11.16
Release Note	HiSIM_SOTB_1.3.0_Release_Note.pdf	2021.11.16
User's Manual	HiSIM_SOTB_1.3.0_Users_Manual.pdf	2021.11.16
Default Model Parameters	HiSIM_SOTB_1.3.0_Default_Model.txt	2021.11.16
Default Instance Parameters	HiSIM_SOTB_1.3.0_Default_Instance.txt	2021.11.16

Update information of HiSIM_SOTB 1.3.0 from HiSIM_SOTB 1.2.0

1. Isub model improvement (Recalculation of Isub after solving the Poisson equations).
2. Improvement of history effect.
3. Implementation of resistance effect (following the implementation in HiSIM_SOI 1.4.0). New instance parameters: **NRS**, **NRD**. New instance and model parameters: **LDRIFTS**, **LDRIFT**. New model flags: **CORS**, **CORD**. New model parameters:

RSH	NOVERS	RDRMUED	RDRMUES	RDRVMAXD	RDRVMAXS
RDRMUETMP	RDRVTMP	RDRDJUNC	RDRBBD	RDRBBS	RDRBBTMP
RDRVMAXW	RDRVMAXWP	RDRVMAXL	RDRVMAXLP	RDRMUEL	RDRMUELP

4. Bugfix of the non-convergence for larger back-gate biases. Fixed by improving the initial potential calculation and by the update of default parameter values (**VBSBND**: 1.0 to 3.0, **VBSMAX**: 1.2 to 3.5).
5. Improvement of capacitance in strong inversion contributed by back gate.
6. Bugfix of the simplified Isub model before solving the Poisson equations (found by Hiroshima University).
7. Bugfix of the negative GDS in the short channel (found by Hiroshima University). Additionally, default parameter values are changed (**CONWUMUB**: 1 to 0, **MUECB0LP**: 1.0 to 0.0, **MUECB1LP**: 1.0 to 0.0, **MUEQB**: 1.0 to 0.0).
8. Bugfix of the induced-gate noise model for the symmetric pins test in CMC-QA, following the bugfix in the other HiSIM models.
9. Code clearing (**COMUUNSZ** is removed).
10. 1/f noise code implementation revised as per the CMC's guidance; resulting in correct Pnoise simulation results.
11. Cgg-Vgs spike issue resolved by mitigating division by zero through applying a correct analytical equation for initial guess
12. Replacement of the loop-flag within the iterative solution of Poisson equation on the source side so that a fall-back solution from analytical equations is adopted.
13. Over-all code clearing with an aid of VAMPyRE 1.8.1 and 1.8.2.